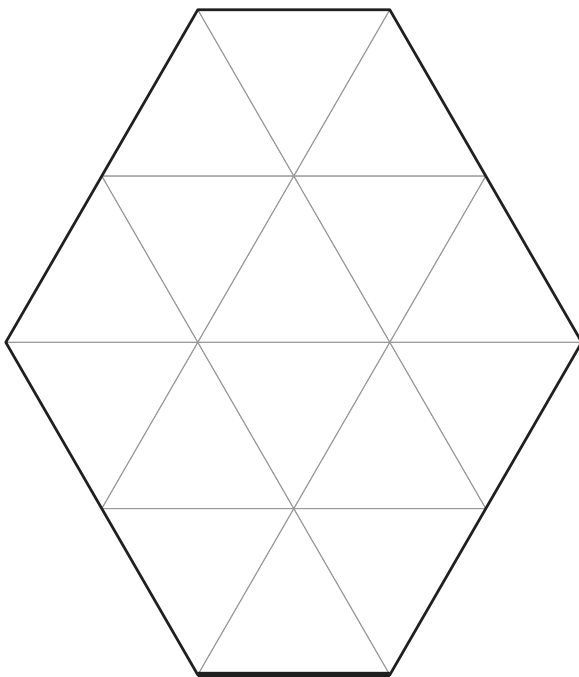


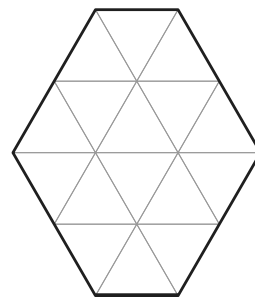
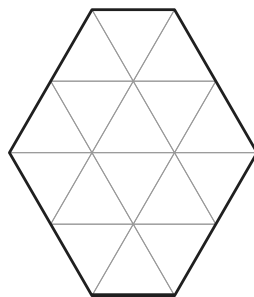
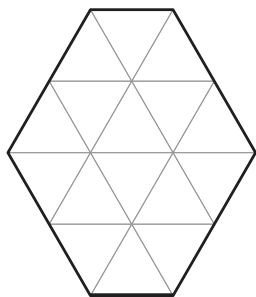
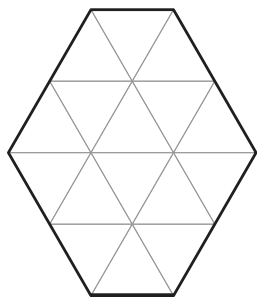
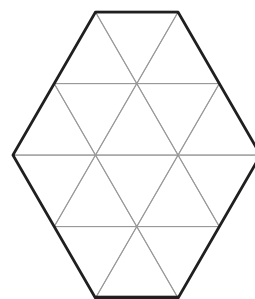
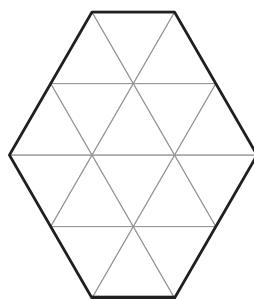
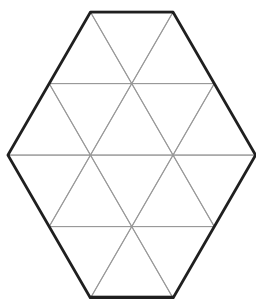
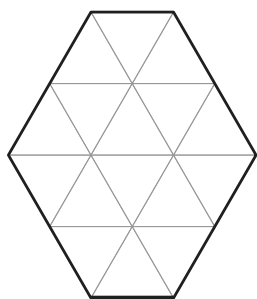
Version W

Week 1 tiling · Grades 4–5

Problem 1: Fill the large board using only small blue rhombi. Find as many different tilings as you can. Keep the bold edge at the bottom. Record each tiling on a small board.

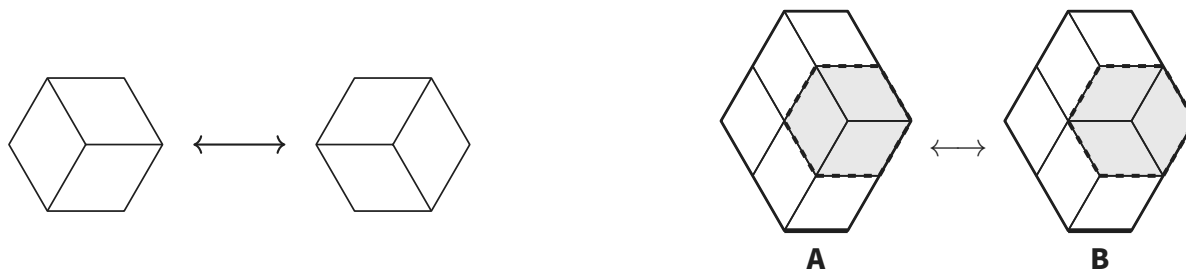


Record each blue rhombus with one line joining the centers of its two triangle spaces.

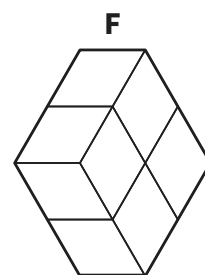
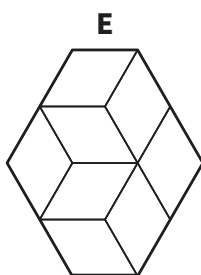
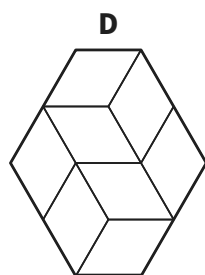
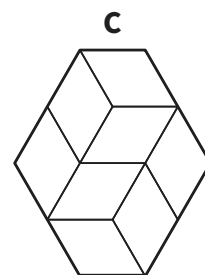
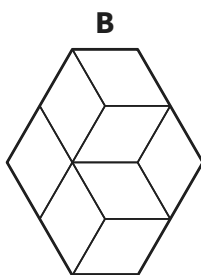
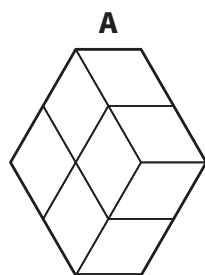


How many different tilings did you find? _____

Problem 2: A flip changes three blue rhombi inside a small hexagon to its other filling. The other rhombi stay in place. Try the flip shown from A to B, then flip back.



Make a map on these six cards. Draw a line between two cards whenever one flip changes one into the other. Check each line with your blocks.



Can you get from every card to every other card by a sequence of flips?

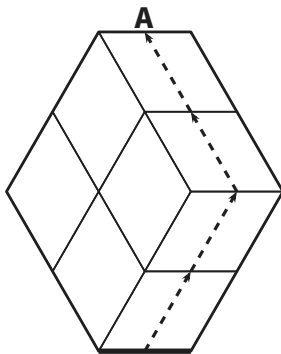
Problem 3: Find the shortest route from A to F. Write the cards you visit and count your flips.

Route: A _____

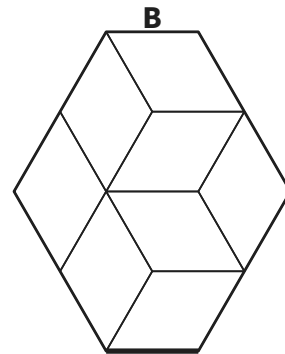
Number of flips: _____

Problem 4: Can you leave A and return to A in exactly three flips? Five flips? Any odd number of flips? Explain what you notice.

Problem 5: Follow a ribbon from the middle of the bold bottom edge. Cross its rhombus to the middle of its opposite horizontal edge. Keep crossing upward until you reach the top. Follow the example on A, then draw the ribbon on B. Write L for each step to the left and R for each step to the right, reading from bottom to top.



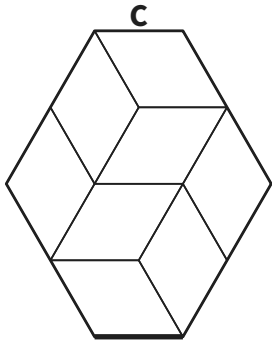
Code: _____



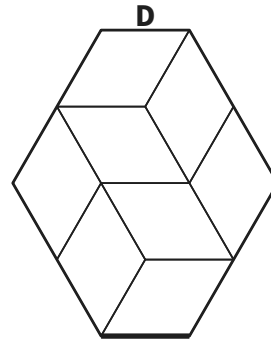
Code: _____

At any step, is there a choice of which rhombus to cross next? Could one tiling have two different ribbons starting at the bold edge?

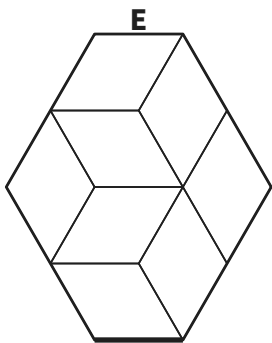
Problem 6: Draw the ribbon on each of these tilings. Write its code from bottom to top.



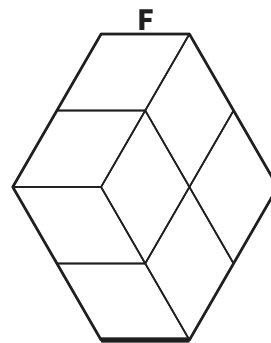
Code: _____



Code: _____



Code: _____



Code: _____

Problem 7: Compare the six ribbon codes from Problems 5 and 6. How many steps does each ribbon take? How many are L? How many are R? Use the shape of the board to explain why.

Write every different order you can make with two L's and two R's. Match each order to a tiling card.

Code: _____

Card: _____

Code: _____

Card: _____

Code: _____

Card: _____

Code: _____

Card: _____

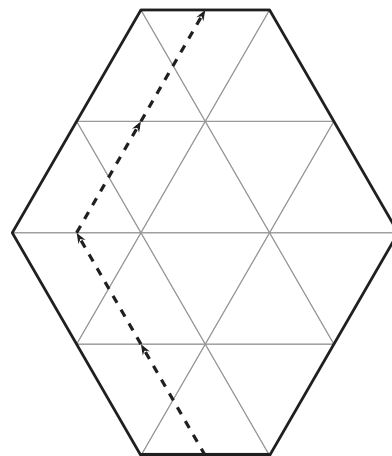
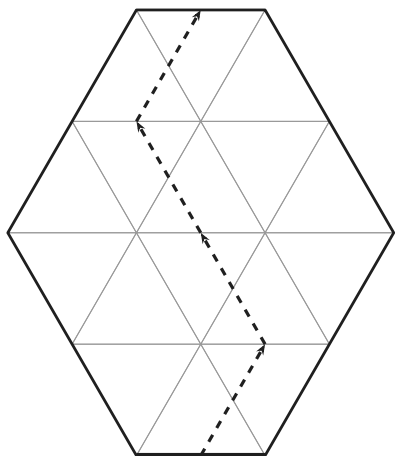
Code: _____

Card: _____

Code: _____

Card: _____

Problem 8: Rebuild the tiling from each ribbon. Draw the four blue rhombi crossed by the ribbon, then fill the remaining spaces with blue rhombi. Can either ribbon give you two different tilings?



Problem 9: Compare the ribbon codes before and after each flip on your map. What changes? Write three examples.

Cards	Code before	Code after
_____	_____	_____
_____	_____	_____
_____	_____	_____

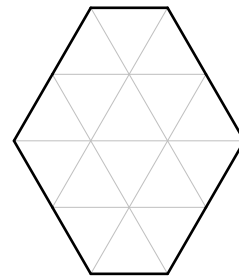
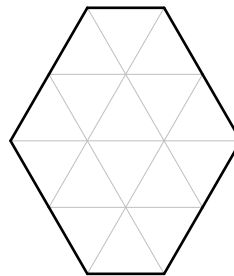
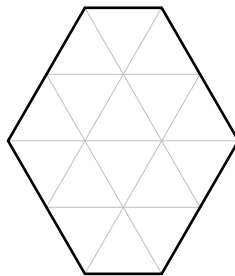
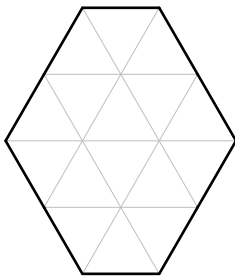
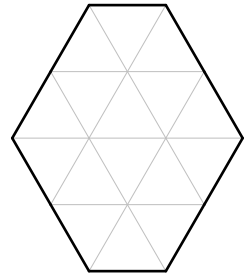
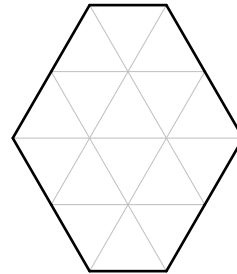
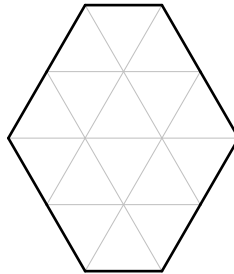
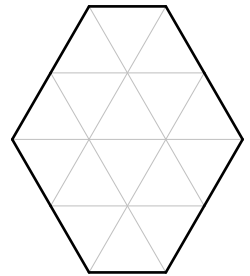
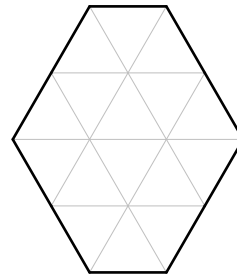
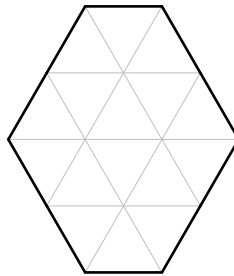
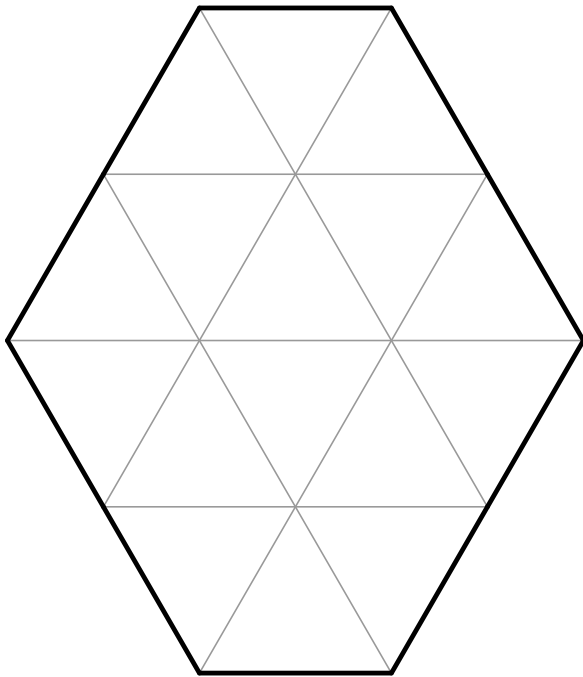
Problem 10: Circle every neighboring RL or LR in your six codes. Can you make a flip at every circled pair? Are there any flips that do not come from one of these pairs? Check with the cards and blocks, then describe your rule.

Version X

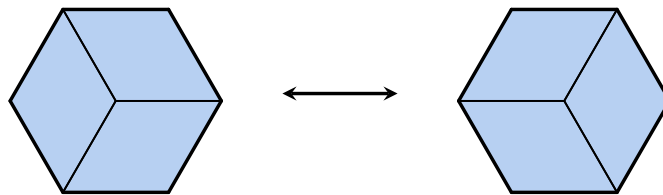
Week 1 tiling · Grades 4–5

On every board in this packet, each block must cover whole small triangles of the board. Blocks may not overlap or stick out past the thick outline.

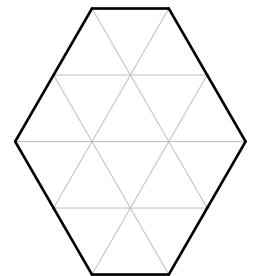
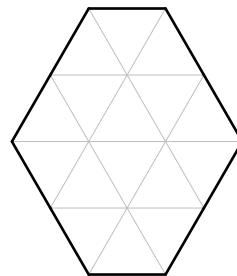
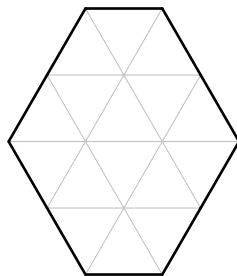
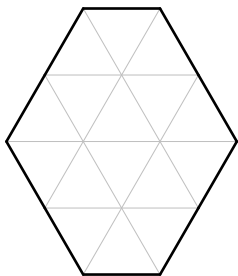
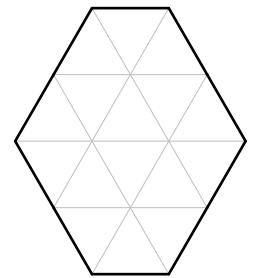
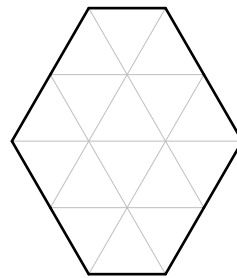
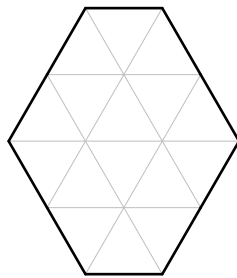
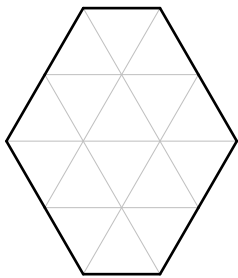
Problem 1: Cover the board below with blue rhombuses . Find all the different ways to do it, and draw each way on one of the small boards. Explain how you know you have found them all.



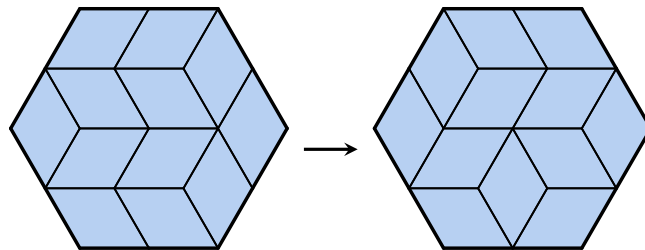
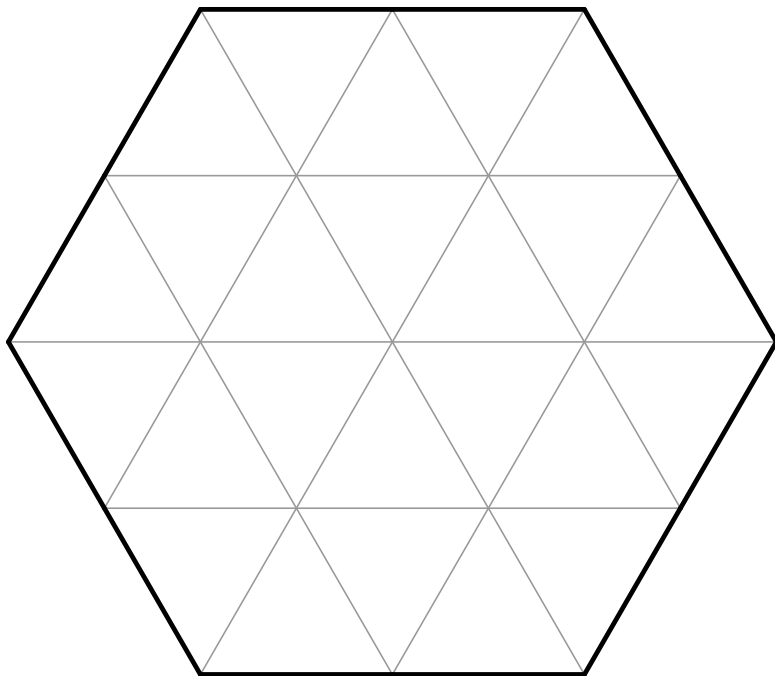
Problem 2: When three blue rhombuses in a covering make a hexagon the size of a yellow block , you can lift them off and put them back the other way, as in the picture. This move is called a flip.



On the small boards below, draw all your coverings from Problem 1 again. Draw a line between two coverings whenever one flip changes one into the other. Which two coverings need the most flips to get from one to the other?

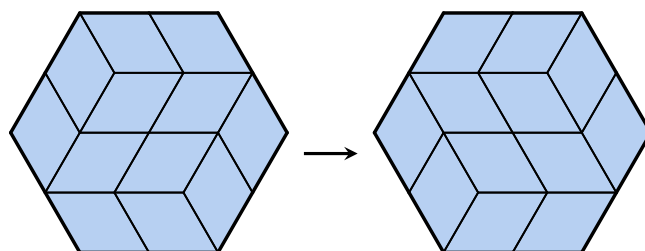


Problem 3: Put blue rhombuses on the hexagon board to make covering A. Change it into covering B using only flips, with as few flips as you can. Do the same for C and D, and for E and F. Write the fewest number of flips you found in each box.



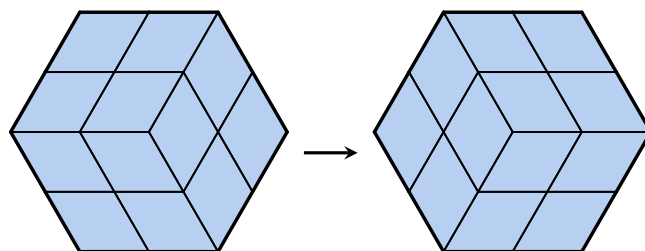
A

B



C

D

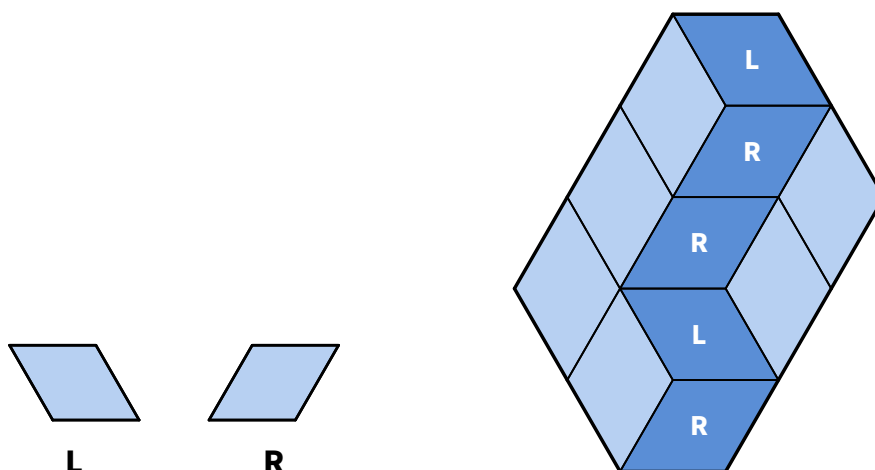


E

F

Problem 4: Start with any covering of the hexagon board from Problem 3. Make flips, one at a time, until you are back at the covering you started with. Find a way to do this with 4 flips that never flips the same hexagon twice in a row. Can you do it with 5 flips? With 7 flips?

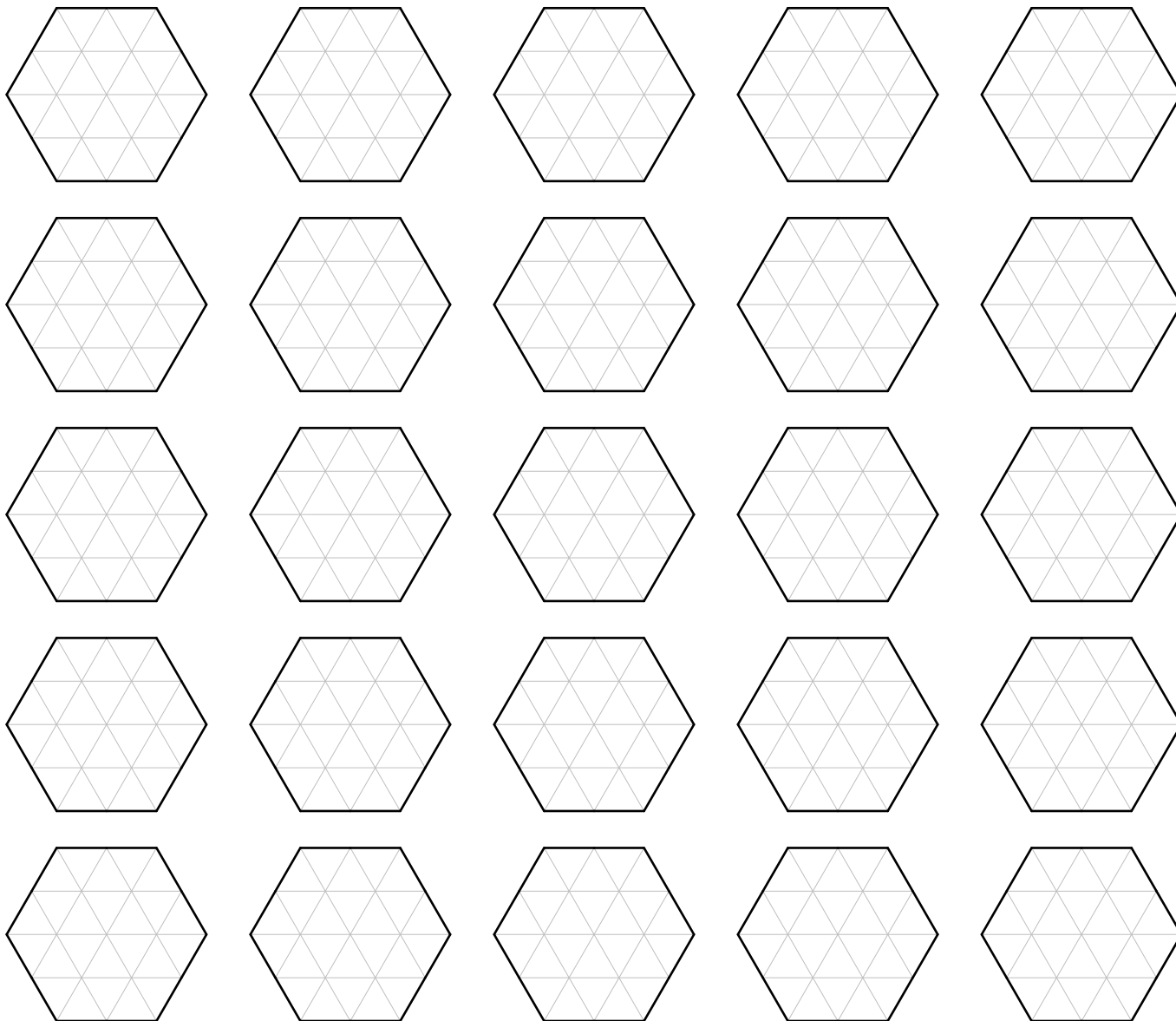
Problem 5: In any covering, pick a short edge on the top side of the board. The rhombus below that edge has a flat bottom edge. Go down to the rhombus below that flat edge, and keep going down this way until you reach the bottom side of the board. Write L for each rhombus you pass through that leans left and R for each one that leans right. The letters, from top to bottom, are the ribbon that starts at that edge. The picture shows a covering of a different board with its ribbon LRRLR.



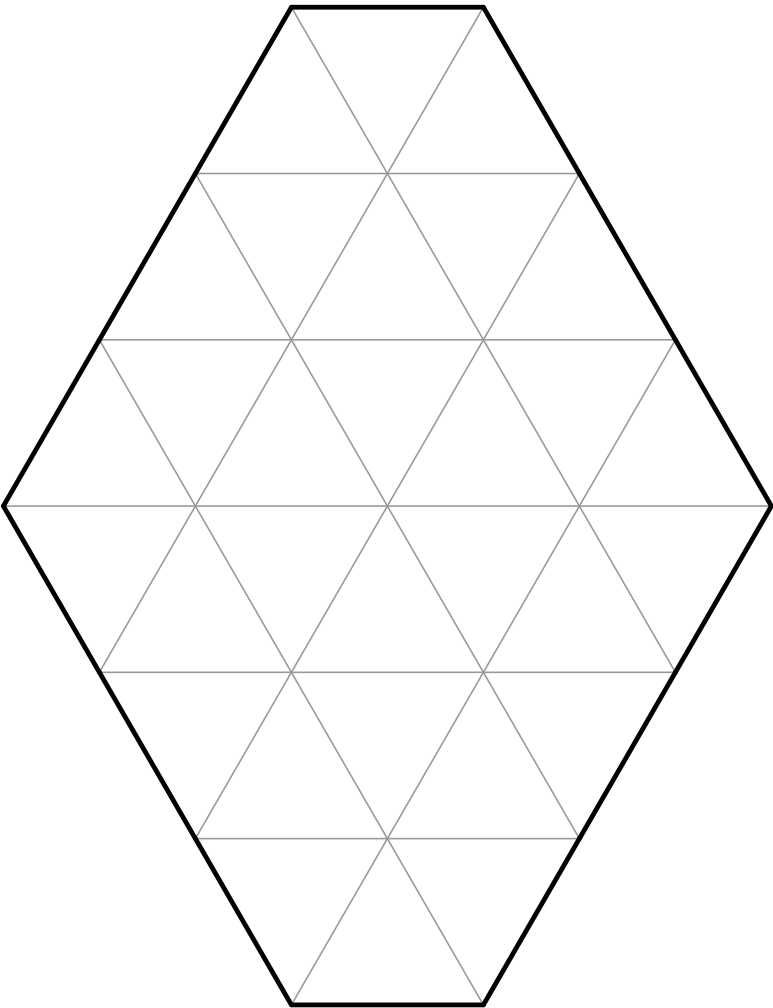
Write the ribbon of each covering of the board from Problem 1. On the hexagon board each covering has two ribbons, because its top side has two short edges. Find exactly what one flip does to the ribbons.

Problem 6: Explain why, whenever you make flips and end at the covering you started with, you have made an even number of flips. Then explain why nobody can change covering E into covering F in Problem 3 with fewer flips than your answer.

Problem 7: How many different coverings does the hexagon board from Problem 3 have? Draw them on the small hexagons below, and explain how you know you have found them all.



Problem 8: How many different ways can blue rhombuses  cover this board? Explain how you know.

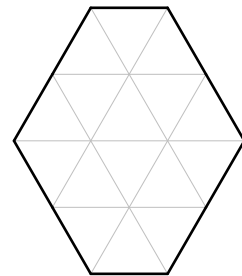
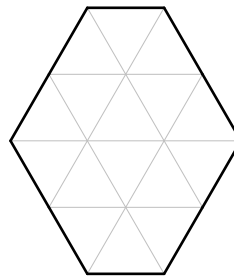
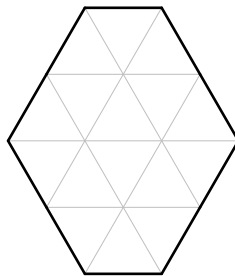
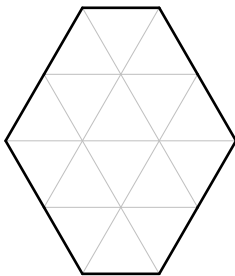
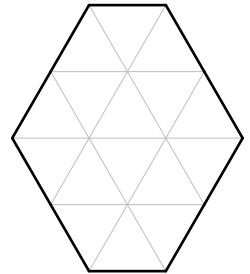
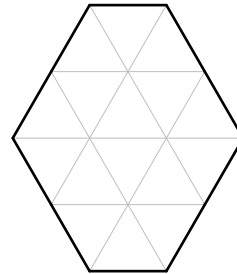
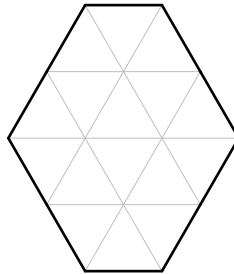
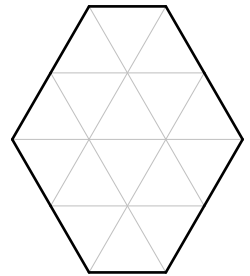
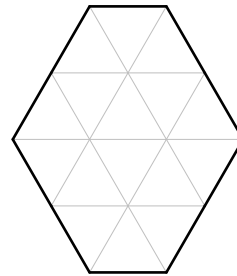
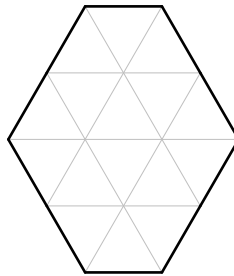
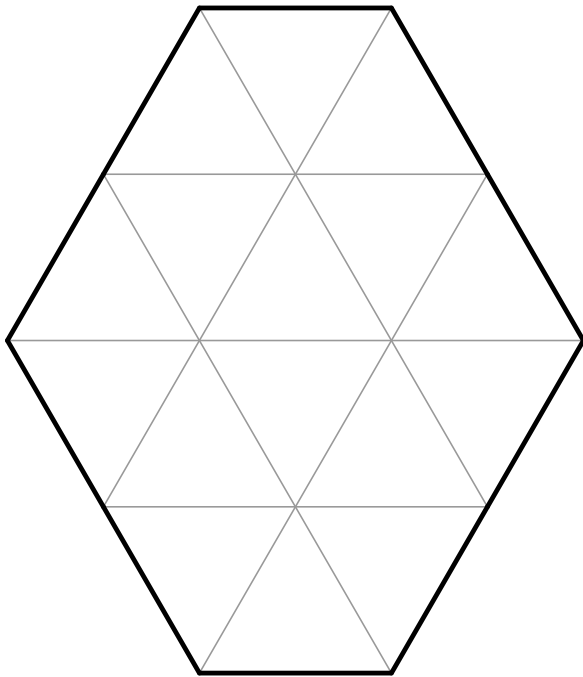


Version Y

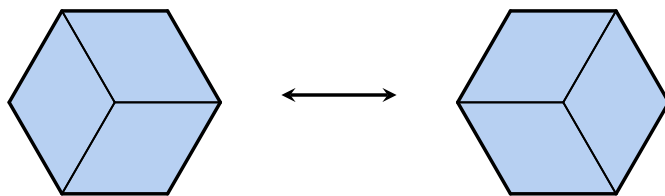
Week 1 tiling · Grades 4–5

Use only the small blocks. On every board in this packet, each block must cover whole small triangles of the board. Blocks may not overlap or stick out past the thick outline.

Problem 1: Cover the board below with blue rhombuses . Find all the different ways to do it, and draw each way on one of the small boards. Explain how you know you have found them all.

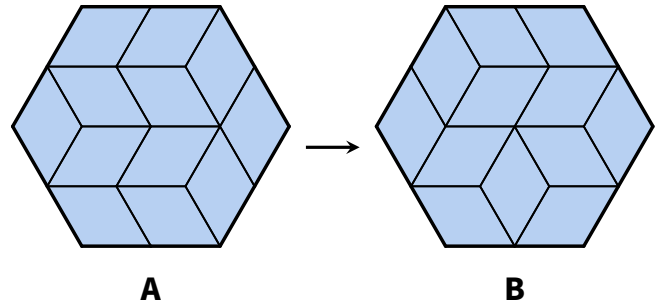
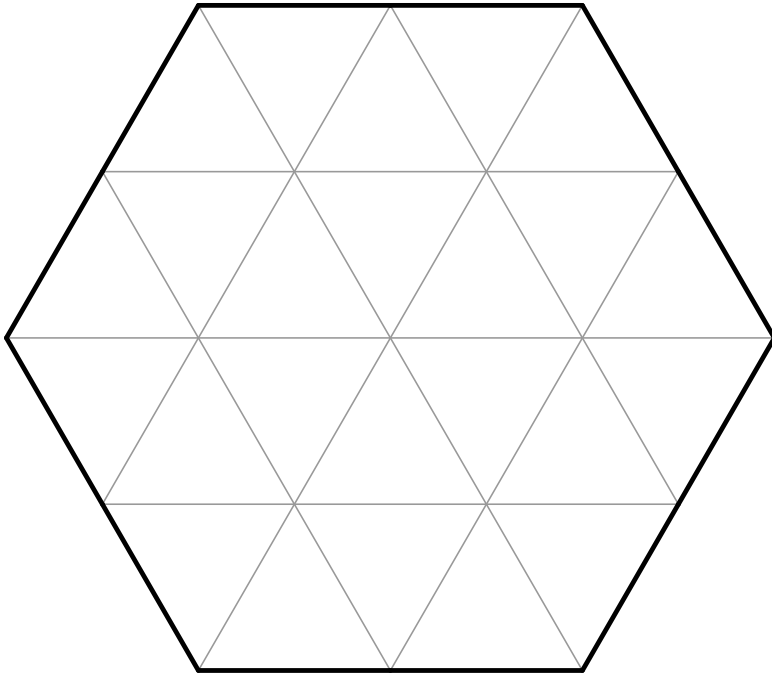


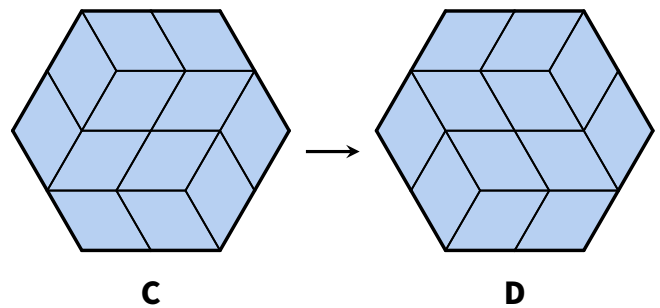
Problem 2: When three blue rhombuses in a covering make a hexagon the size of a yellow block , you can lift them off and put them back the other way, as in the picture.

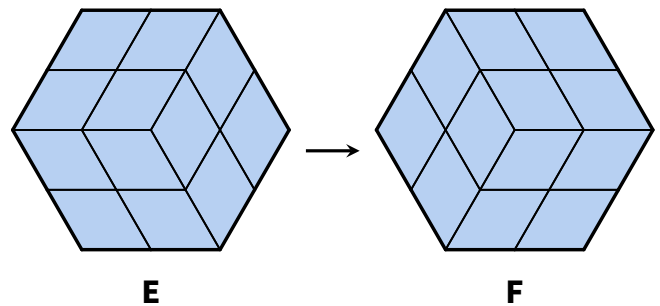


Use this move to change one covering of the board from Problem 1 into another. Which two coverings need the most moves to get from one to the other? How many moves do they need?

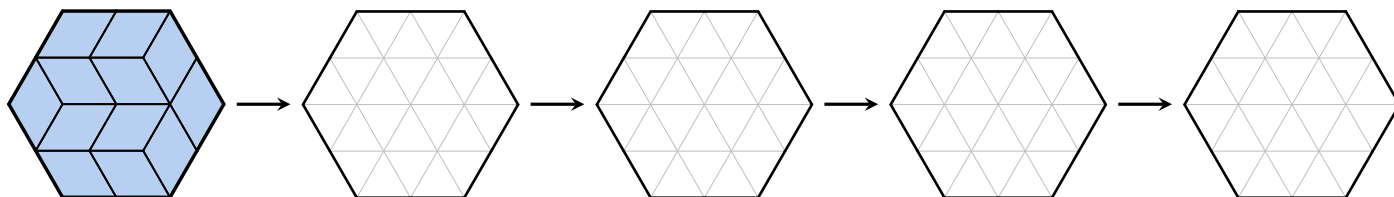
Problem 3: The move in Problem 2 is called a flip. Put blue rhombuses on the hexagon board to make covering A. Change it into covering B using only flips, with as few flips as you can. Do the same for C and D, and for E and F. Write the smallest number of flips you found in each box.



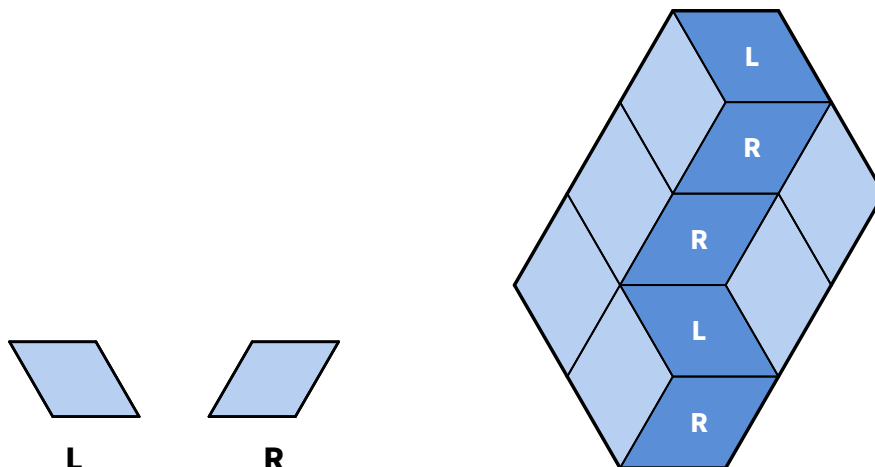




Problem 4: Make covering A from Problem 3 on the hexagon board. Make flips, one at a time, until you are back at covering A. Find a way to do this with 4 flips that never flips the same hexagon twice in a row. Draw the covering after each flip on the small hexagons. Can you get back to covering A with 5 flips? With 7 flips?



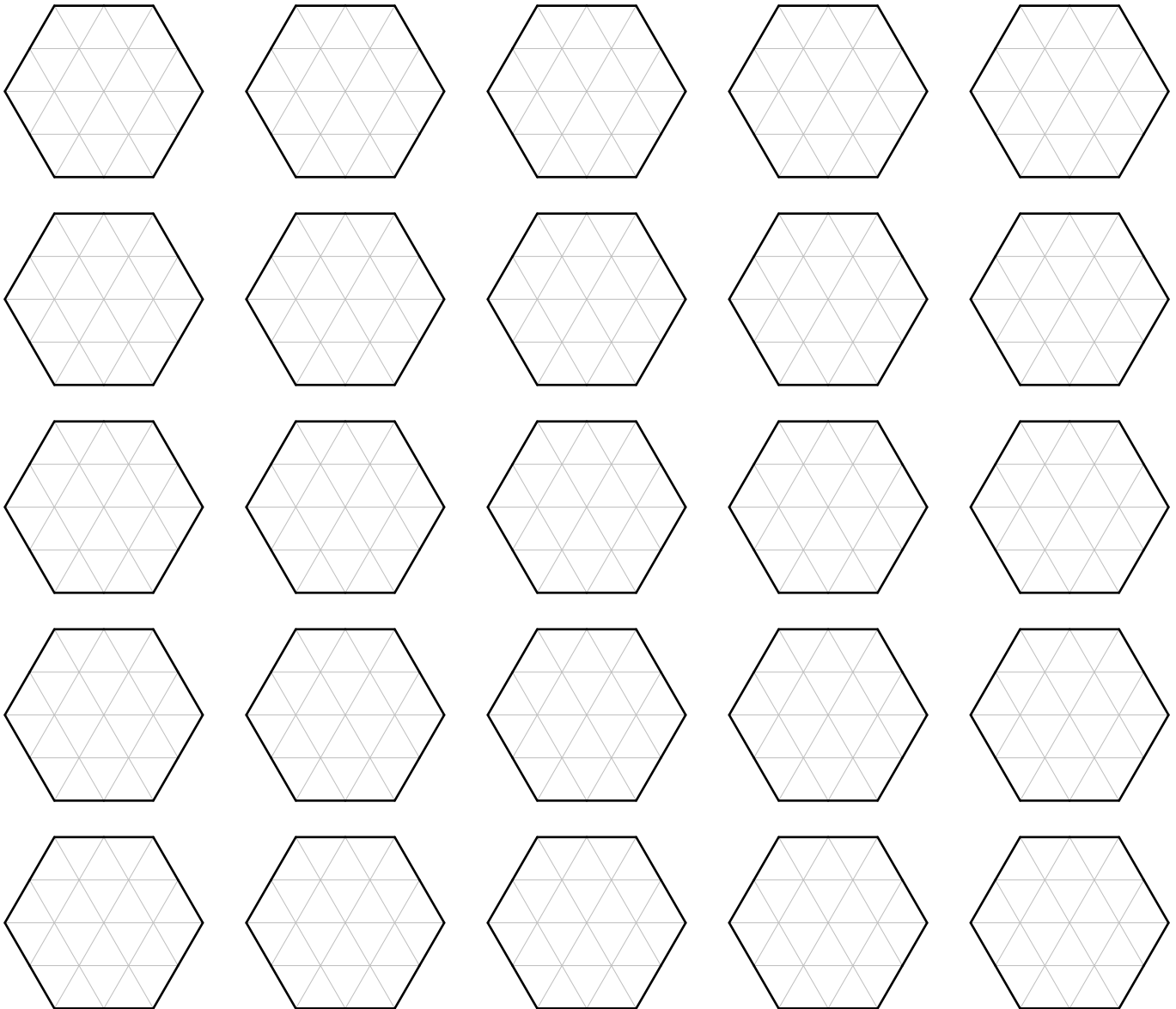
Problem 5: In a covered board, start at a short edge on the top side. Go down through the rhombuses, from flat edge to flat edge, until you reach the bottom side. Write L for each rhombus you pass that leans left and R for each one that leans right. The picture shows a covering of a different board, where going down gives LRRLR.



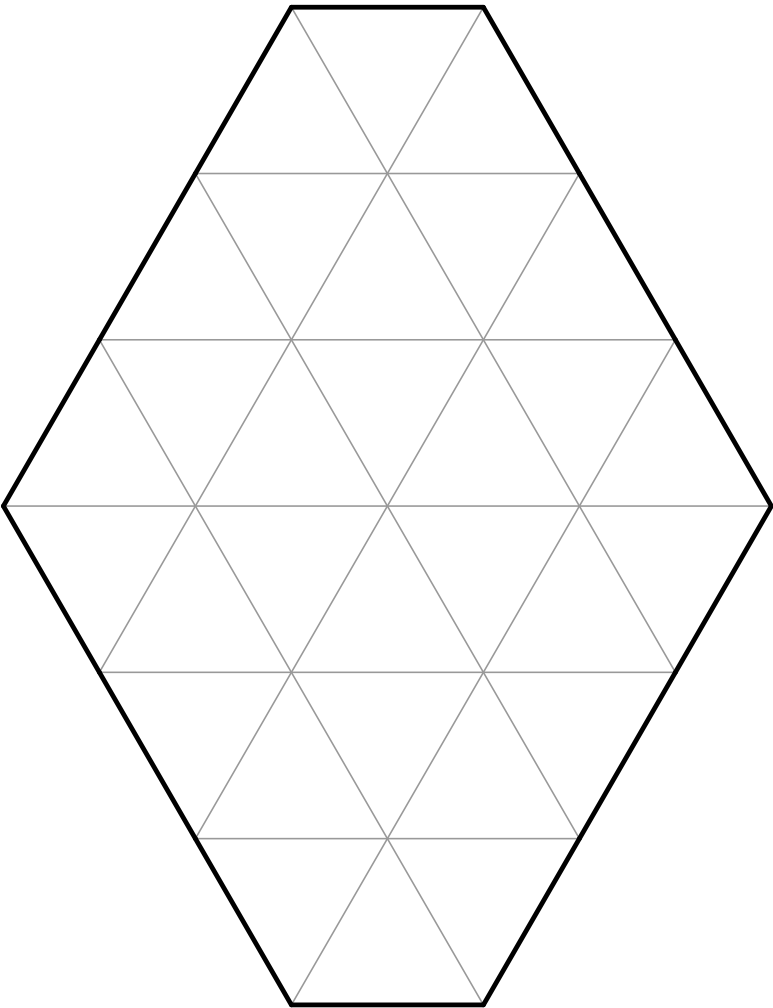
Write the letters for each covering of the board from Problem 1. On the hexagon board from Problem 3, the top side has two short edges, so each covering gets two strings of letters. Call them its ribbons. Find exactly what one flip does to the ribbons.

Problem 6: Explain why, whenever you make flips on the hexagon board and end at the covering you started with, you have made an even number of flips. Then explain why nobody can change covering E into covering F in Problem 3 with fewer flips than your answer.

Problem 7: How many different coverings does the hexagon board from Problem 3 have? Draw or list them below, and explain how you know you have found them all.



Problem 8: How many different ways can blue rhombuses  cover this board? Explain how you know.



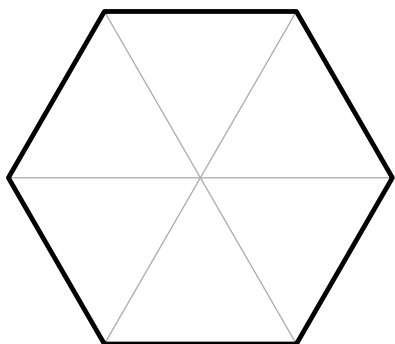
Version Z

Week 1 tiling · Grades 4–5

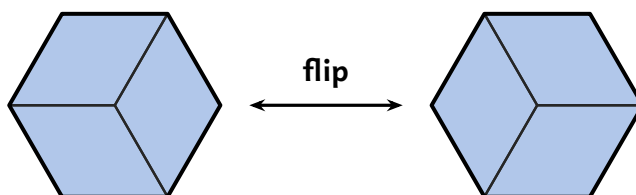
1 Three blues, two ways

Name: _____

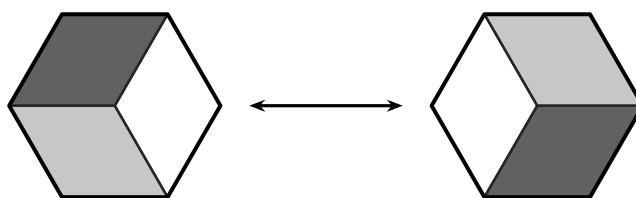
Today we only use **blue rhombi**. On our grid a blue can sit in three directions:



1. Cover this hexagon with 3 blues. There are exactly **two** ways. Build both.
2. Lift the three blues and turn them, so that one way becomes the other. This move is called a **flip**.

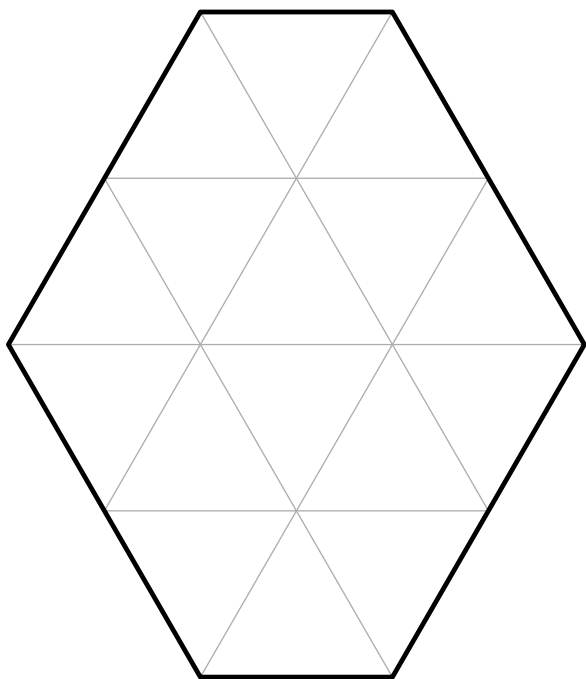


3. **Look in 3D.** Here are the same two tilings, shaded: one direction white, one light gray, one dark gray. Stare at them. What do you see? What does a flip do to the picture?



Coming up. Bigger hexagons have many tilings. You will hunt for all of them, connect them by flips, and find a secret code hiding inside every tiling.

2 The tall hexagon

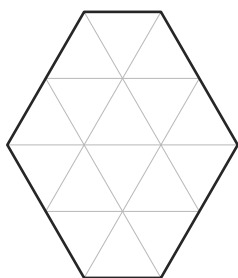


Cover this board with 8 blue rhombi. This is called a **tiling**.

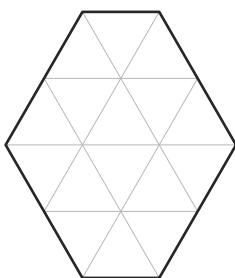
How many *different* tilings can your group find? Draw each new one in a box below and give it a letter.

Tips.

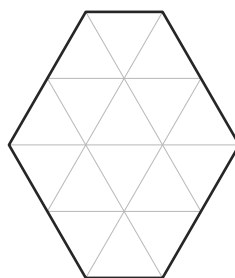
- One person builds, one checks that it is new, one draws.
- Stuck? Look for three blues that make a small hexagon, and flip them!



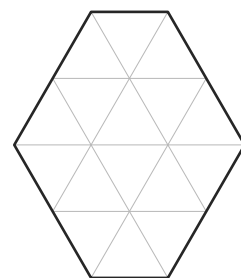
A



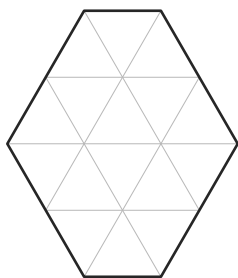
B



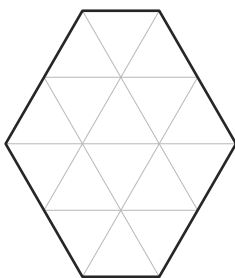
C



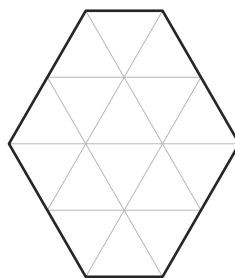
D



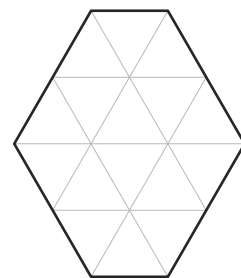
E



F



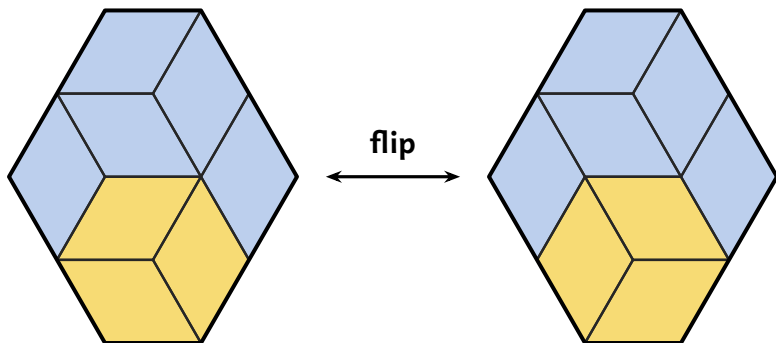
G



H

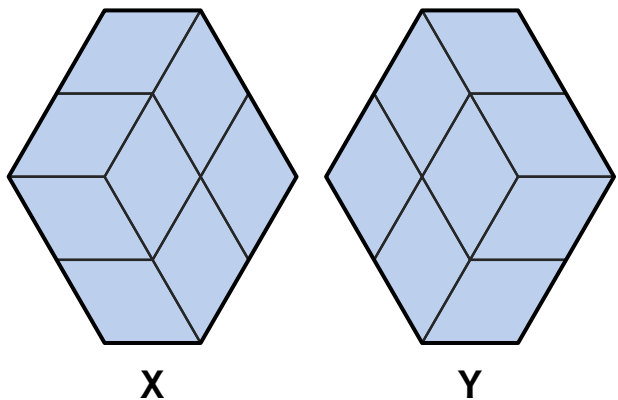
We found _____ tilings. Are you sure there are no more? Why?

3 Flips



Wherever three blues make a small hexagon (shaded yellow), you can flip them. Two tilings are **neighbors** if one flip changes one into the other.

1. Flip map. Write the letters of your tilings from page 2 in this box. Draw a line between every pair of neighbors. Move letters around until your map looks tidy.

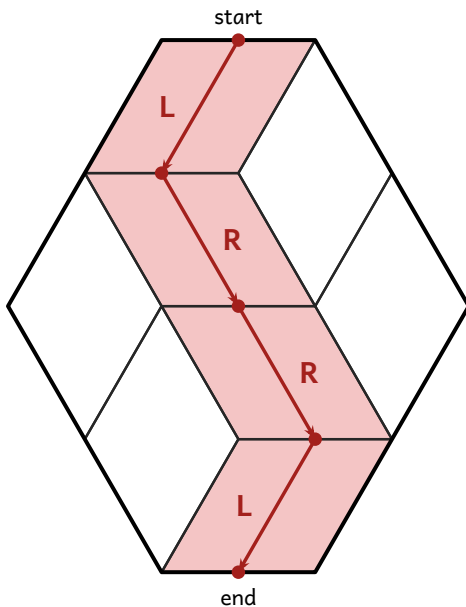


2. What is the fewest number of flips that turns X into Y? _____

How do you know it can't be done in fewer?

3. Round trips. Start at a tiling, make flips, and come back to where you started. Count the flips. Can you find a round trip with an *odd* number of flips?

4 Ribbons: a secret code



Start at the **top edge**. The blue touching it has a matching edge at its bottom. The next blue shares that edge. Keep going down to the bottom edge. This chain of blues is a **ribbon**.

Each step of a ribbon goes down-left (**L**) or down-right (**R**). The ribbon in the picture spells **LRRL**.

1. Shade the ribbon in each of your tilings from page 2 and write its word.

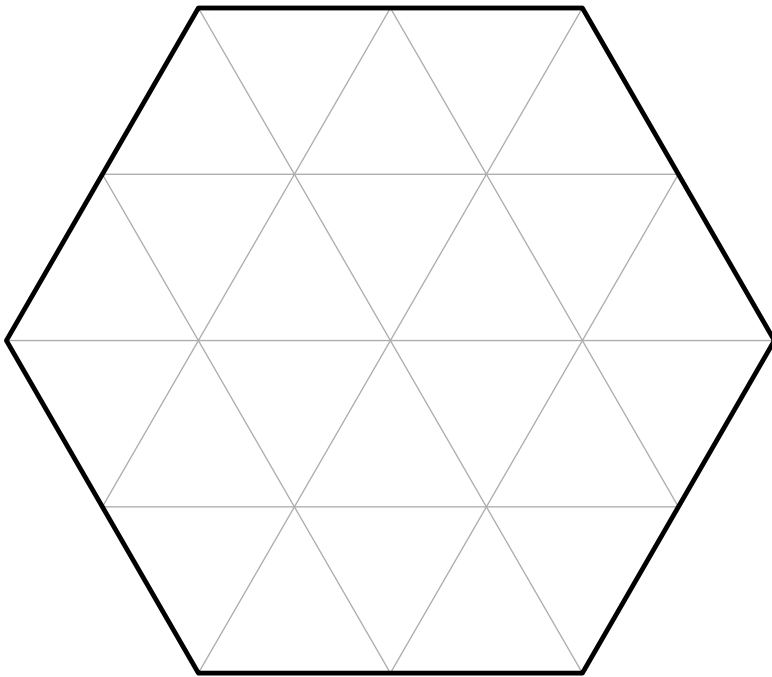
Tiling	A	B	C	D	E	F	G	H
Word								

2. List *every* word made of two L's and two R's. Does each word give a tiling? So how many tilings does the tall hexagon have?

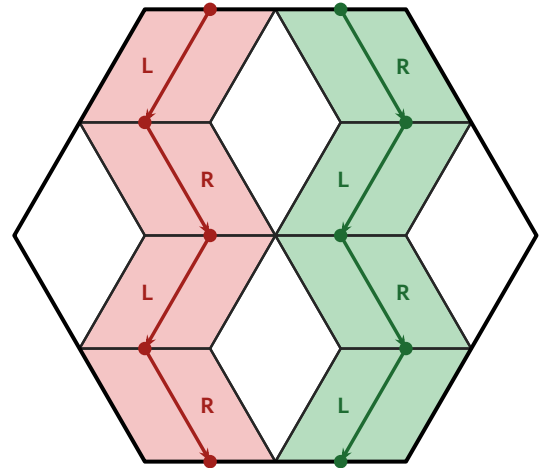
3. Pick two neighbors from your flip map and compare their words. What does one flip do to the word?

4. The **L-score** of a word is the sum of the places where the L's are. For LRRL it is $1 + 4 = 5$. What does one flip do to the L-score? Use this to explain why a round trip can *never* have an odd number of flips.

5 The big hexagon



Tile this board with 12 blues.
Now **two** ribbons start at the top edge.



Left ribbon: L R L R

Right ribbon: R L R L

1. Build a tiling and find both ribbons.

Left ribbon: _____ Right ribbon: _____

2. Can the two ribbons cross? Can they share a blue? Which pairs of words can go together? (Your flip map from page 3 can help!)

3. **Challenge.** How many tilings does the big hexagon have? Explain how you counted. Keep track on paper or a whiteboard.

4. **Look in 3D.** Shade one of your tilings like the picture. What do you see? What does a flip do to it?

